

Please check the examination details below before entering your candidate information

Candidate surname

Other names

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Centre Number

Candidate Number

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Pearson Edexcel International GCSE (9–1)

By Enidu Batuwanthudawe

Afternoon (Time: 1 hour 30 minutes)

Paper
reference

4IT1/01

Information and Communication

Technology (ICT)

PAPER 1: Written Paper

You do not need any other materials.

Total Marks

Instructions

- Use **black** ink or ball-point pen.
- **Fill in the boxes** at the top of this page with your name, center number and candidate number.
- Answer **all** questions.
- Answer the questions in the spaces provided
 - *there may be more space than you need.*

Information

- The total mark for this paper is 100.
- The marks for **each** question are shown in brackets
 - *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

Turn over ►

Answer ALL questions.

Write your answers in the spaces provided.

Some questions must be answered with a cross in a box ☒. If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒.

Yash hosts a live interactive radio stream using professional audio equipment



(Source:CircleBit.net)

1 Yash streams an online live radio show.

- (a) State one **specialized input device** (other than a basic microphone) that could be used to capture Yash's voice, and give one reason why it is suitable. (2)

- (b) Yash needs to minimize latency and ensure high-quality audio transmission between his input device and computer.

- (i) Which one of these is the most suitable connectivity method for this scenario? (1)

- ☐ A Bluetooth
- ☐ B NFC
- ☐ C USB (wired)
- ☐ D Wi-Fi

- (ii) Explain why your chosen answer in (b)(i) is more suitable than one other option. (2)

- (iii) Which one of these best describes the impact of high latency on a live stream?

- ☐ A Audio becomes delayed and out of sync with real-time events
- ☐ B Audio file size increases
- ☐ C Volume levels automatically increase
- ☐ D Sampling rate decreases

- (c) Describe in detail how streaming works when Yash broadcasts her radio show. (3)

(d) Yash wants to stream her show globally and allow listeners to join live.
State the type of network used and explain why it is required instead of a LAN (2)

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(e) Explain one way copyright legislation affects how a listener can use a recording of Yash’s live radio stream. (2)

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(f) **Yash’s computer has 16 GiB of RAM.**
Each uncompressed recording uses **512 MiB of RAM.**

(i) Construct an **expression** to show how many recordings can be stored in RAM at once. (2)

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(ii) Which one of these is a benefit to Yash of increasing RAM? (1)

- ☐ **A** More recordings can be processed simultaneously without slowing down
- ☐ **B** The internet speed will increase automatically
- ☐ **C** Audio files will become compressed automatically
- ☐ **D** The operating system will require less storage space

(iii) State two benefits of using secondary storage, rather than RAM, to save the radio recordings. (2)

1).....

2).....

(iv) Which one of these does the 'A' stand for in RAM?

(1)

- ☐ **A** Access
- ☐ **B** Allocation
- ☐ **C** Array
- ☐ **D** Available

(v) State the reason why RAM is described as 'random access'.

(2)

(g) **The audio data is processed.**

(i) State how the speed of a processor is measured.

(1)

(i) State one feature of a laptop that allows expansion or improvement of audio processing performance.

(1)

(h) **Listeners can subscribe to receive updates about new shows.**

(i) State one risk to a listener's personal information when subscribing online.

(1)

(ii) Describe one way a listener can protect their personal information when subscribing.

(2)

(iii) There are legal requirements for storing personal data.

Explain two legal requirements Yash must follow when storing listeners' personal information.

(4)

1).....

2).....
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(iv) Users who subscribe are shown recommended content. (2)
Describe how an online service can provide the **most relevant content** to its users.

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2 Yash shares links to her live radio streams on an online platform.

(a) Explain one reason why Yash avoids uploading high-resolution video clips while streaming live. (2)

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(b) Which one of these is the most suitable type of online community for Yash to promote and discuss her shows? (1)

- ☐ A Online gaming platform
- ☐ B Social networking site
- ☐ C Virtual Learning Environment (VLE)
- ☐ D Wiki

(c) Anonymity can create risks in online communities.

(i) State one other threat to users in online communities. (1)

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(ii) Describe one effective method moderators can use to improve safety in online communities. (2)

(iii) State the purpose of an acceptable use policy (AUP) in an online community. (1)

(d) Describe how one feature of a social networking platform can help create and maintain online communities. (2)

(e) People search for information about Yash's shows online.

(i) Which one of these would most negatively affect the reliability and usefulness of information found online? (1)

- ☐ **A** Bias
- ☐ **B** Collaboration
- ☐ **C** Encryption
- ☐ **D** Referencing

(ii) State two methods a user can use to check the reliability of online information. (2)

1).....

2).....

(iii) Which one of these is the best way to avoid plagiarism when using online information? (1)

- ☐ A Copying directly from a website
- ☐ B Using copyright symbols
- ☐ C Paraphrasing and referencing sources
- ☐ D Using only one source

(f) Describe one-way features of an online community can increase the reach of Yash’s posts. (2)

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(g) Which one of these actions best improves a user’s online safety? (1)

- ☐ A Sharing personal location publicly
- ☐ B Accepting all friend requests
- ☐ C Limiting the amount of personal information shared
- ☐ D Posting frequently on public forums

3 Yash connects to different types of networks to broadcast her radio show.

(a) **Yash connects to the Internet**

(i) State the main purpose of an Internet Service Provider (ISP). (1)

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(ii) Yash connects a laptop to a wireless access point.

State two additional components required in a network to allow Internet access. (2)

1).....

2).....

(iii) Describe the role of a web browser when accessing content on the Internet. (2)

(iv) State one use of an IP address in a network. (1)

(v) Explain one advantage of using a MAC address instead of an IP address for device identification. (2)

(b) **Yash uses networks to stream her show.**

(i) Give two types of wireless connectivity Yash could use to connect a laptop to a smartphone. (2)

- 1).....
- 2).....

(ii) Explain one reason why a WAN is required instead of a LAN for streaming to a global audience. (2)

(c) **Yash uses GPS features.**

(i) Which one of these statements about GPS is correct? (1)

- ☐ **A** Devices send data to satellites to calculate position
- ☐ **B** GPS requires connection to a mobile network
- ☐ **C** GPS receivers calculate position using signals from multiple satellites
- ☐ **D** GPS only works indoors

(ii) Which one of these is a valid use of GPS in Yash's context? (1)

- ☐ **A** To increase Internet speed
- ☐ **B** To embed location metadata in uploaded media
- ☐ **C** To encrypt transmitted data
- ☐ **D** To replace Wi-Fi connectivity

(d) Describe one role of a server in a client-server network used by Yash. (2)

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(e) Explain one advantage of using a client-server network instead of a peer-to-peer network. (2)

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(f) Explain one benefit to Yash of using a VPN when connecting to the radio station's network remotely. (2)

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ENIDU BATUWANTHUDAWA

DO NOT WRITE IN THIS AREA

WWW.EDEXCEL.COLLEGE

(g) Yash's wired broadband connection fails, but her smartphone still has mobile data.

Describe one method Yash could use to keep her laptop connected to the Internet. (2)

(h) Yash tests her connection:

- **Download speed:** 25 Mbps
- **Upload speed:** 0.05 Mbps

Explain the impact of these speeds on the experience of Yash's live audience. (2)

4 Yash uses a laptop to record and manage high-quality audio.

(a) Which one of these would most appropriately represent the **minimum practical amount of data** required to store a one-hour **uncompressed audio recording**? (1)

- ☐ A Byte
- ☐ B Kibibyte
- ☐ C Mebibyte
- ☐ D Gibibyte

(b) State one operating system function that is essential during audio recording. (1)

(c) Explain one benefit to Yash of using **open-source audio recording software** instead of proprietary software. (2)

(d) Explain two reasons why Yash should regularly update her audio recording software and operating system. (2)

1).....

2).....

(e) Yash travels to different locations and carries a large music and recording collection.

Discuss the benefits and drawbacks of different storage media he could use.

You should include in your answer:

- different types of storage (e.g. SSD, HDD, cloud storage, USB flash)
- portability
- capacity
- compatibility

durability/reliability

(8)

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5 Yash’s use of digital technologies has wider implications.

(a) State two causes of the digital divide. (1)

1).....

2).....

(b) Explain two negative **social impacts** of limited access to the Internet. (4)

1).....

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2).....

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(c) Yash’s radio show is streamed online.

Explain two negative **environmental impacts** of streaming and storing digital media online (4)

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2).....

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(d) Yash's show is available both as a traditional radio broadcast and online

Discuss how the reliability of broadcast information compares with online information.

You should include:

- regulation and control of broadcast media
- user responsibility when accessing online information
- methods used to verify accuracy of information

(8)

Handwriting practice area with 20 horizontal dotted lines.